

Camera Slider

12864 Screen Mod — Wiring & Connection Reference
Firmware v1.0-12864

Component	Part	Interface / Notes
Microcontroller	Arduino Uno	—
Display	BTT Mini12864 V2.0	SPI (software), EXP1+EXP2
Encoder	Built into Mini12864	EXP2-3/5, EXP1-2
Stepper Driver	TMC2209	STEP/DIR — EN tied to GND
Stepper Motor	NEMA 17	4-wire to TMC2209
RGB Backlight	WS2812 (on Mini12864)	A4 via EXP1-7
Endstops	2× NC switch — parallel	Single pin D12
IR Receiver	Standard module	D7 (NEC protocol)
LEFT button	Panel mount momentary	A0
RIGHT button	Panel mount momentary	D5
PLAY/PAUSE button	Panel mount momentary	D6
E-Stop	Encoder long press	Built-in — no extra button

1 Power Supply

Rail	Source	Connects To	Notes
5V (logic)	USB or Uno 5V	Mini12864 EXP1-10, IR VCC	Logic supply
12–24V (VM)	External PSU	TMC2209 VM	Motor supply
GND	Common	All + TMC2209 EN	EN tied to GND permanently

TMC2209 EN is wired directly to GND — motor always enabled. Coils stay energized at rest, providing holding torque.

2 Mini12864 EXP1 Connector (10-pin IDC)

EXP1 Pin	Signal	Arduino Pin	Function
1	BEEPER	—	Not connected
2	BTN_ENC	D4	Encoder push — short=select, long=E-Stop
3	LCD_RST	D8	Display reset
4	LCD_A0/RS	A3	Data / command select
5	BTN_BACK	—	Not connected (Back is a menu item)
6	BTN_KILL	—	Not connected (PLAY/PAUSE is external)
7	NEOPIXEL	A4	WS2812 RGB backlight data
8	GND	GND	Ground
9	KILL	—	Not connected
10	5V	5V	Logic supply to Mini12864

3 Mini12864 EXP2 Connector (10-pin IDC)

EXP2 Pin	Signal	Arduino Pin	Function
1	MISO	—	Not connected (software SPI — MISO unused)
2	SCK	D13	SPI clock
3	ENC_CLK	D2	Encoder channel A (hardware interrupt)
4	SD_CS	—	Not connected (no SD card)
5	ENC_DT	D3	Encoder channel B
6	MOSI	D11	SPI data to display
7	LCD_CS	A5	Display chip select
8	GND	GND	Ground
9	NC	—	Not connected
10	5V	—	Not connected (powered via EXP1-10)

4 Encoder & Button Behaviour

Control	Short Press / Turn	Long Press (>400 ms)
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Encoder turn	Scroll menu / adjust value	—
Encoder push (D4)	Select / confirm	E-Stop — decel + disable + main menu
LEFT button (A0)	1 mm slow step left	Fast continuous jog left
RIGHT button (D5)	1 mm slow step right	Fast continuous jog right
PLAY/PAUSE (D6)	Start / pause / resume	Cancel active program

All external buttons wired between pin and GND. Arduino INPUT_PULLUP provides pull-up — no external resistor needed.

5 Endstops (NC — both in parallel on one pin)

Connection	Detail
Left endstop	One leg → D12, other leg → GND
Right endstop	One leg → D12, other leg → GND (parallel with left)
Direction logic	speed < 0 → moving left → left endstop triggered
	speed > 0 → moving right → right endstop triggered

NC (Normally Closed) wiring: circuit closed at rest, opens on trigger. A broken wire also triggers the stop — safer than NO wiring.

6 TMC2209 Stepper Driver

TMC2209 Pin	Arduino / Source	Notes
STEP	D9	Step pulse — one per microstep
DIR	D10	Direction
EN	GND (direct)	Tied LOW permanently — always enabled
VIO	5V	Logic supply
VM	12–24V PSU	Motor power
GND	GND (both)	Logic and power GND joined at PSU
A1/M1, A2/M2	Motor coil A+, A-	NEMA 17 coil A
B1/M3, B2/M4	Motor coil B+, B-	NEMA 17 coil B

If motor runs backwards, swap A+ and A- (or B+ and B-). MS1/MS2 pads set microstepping — default 16μstep = STEPS_PER_MM 80.

7 IR Receiver Module

IR Pin	Arduino	Notes
DATA	D7	Signal output — NEC protocol
VCC	5V	
GND	GND	

Upload IR_Learn.ino, open Serial Monitor at 115200 baud, press each button to read its hex code. Paste into IR_BTN_* defines in 12864-screen-mod.ino.

8 NeoPixel RGB Backlight (on Mini12864 board)

State	Colour	R	G	B
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Idle / menu	Green	0	180	0
Program running	Blue	0	0	180
Paused	Amber	200	120	0
Homing	White	200	200	200
E-Stop	Red	255	0	0

9 Arduino Uno — Complete Pin Summary

Pin	Dir	Connected To	Function
D2	In	EXP2-3 ENC_CLK	Encoder A (hardware interrupt)
D3	In	EXP2-5 ENC_DT	Encoder B
D4	In	EXP1-2 ENC_SW	Encoder push (select / E-Stop)
D5	In	RIGHT jog button	Short=step right, hold=fast right
D6	In	PLAY/PAUSE button	Short=start/pause, hold=cancel
D7	In	IR receiver DATA	IR remote input
D8	Out	EXP1-3 LCD_RST	Display reset
D9	Out	TMC2209 STEP	Stepper step pulse
D10	Out	TMC2209 DIR	Stepper direction
D11	Out	EXP2-6 MOSI	SPI data (software SPI)
D12	In	Both endstops (NC parallel)	Single endstop pin
D13	Out	EXP2-2 SCK	SPI clock (software SPI)
A0	In	LEFT jog button	Short=step left, hold=fast left
A1	—	Free	Available for future use
A2	—	Free	Available for future use
A3	Out	EXP1-4 LCD_A0	Display RS
A4	Out	EXP1-7 NeoPixel	RGB backlight data
A5	Out	EXP2-7 LCD_CS	Display chip select
5V	Pwr	EXP1-10, IR VCC	Logic supply
GND	Pwr	All + TMC2209 EN	Common ground + EN tie-down